1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, the core puzzle is why some bladder cancers fail to benefit from intravesical BCG despite the therapy's capacity to elicit local immune responses — specifically, whether local tissue programs (e.g., HOXD13-associated remodeling) shape whether BCG elicits productive lymphoid organization (TLS) or a non-productive, myeloid-dominant state. 2) Tier 1 paths used: I built primarily on Core Path 1 (BCG as the canonical therapy for bladder cancer, providing the clinical intervention context) and Core Path 2 (immune cell heterogeneity; the lymphoid vs myeloid balance as the mechanistic determinant of BCG efficacy). I incorporated Core Path 3 (HOXD13 upregulation in tumor-adjacent tissue) as the molecular anchor that could plausibly drive stromal/tissue remodeling and thereby modulate immune organization. These three together form a mechanistic chain: HOXD13 → tissue remodeling/invasion programs → altered immune composition/spatial organization → modified BCG efficacy. 3) Tier 2 supporting evidence: The supporting paths (risk-class differences in immunotherapy responsiveness, tumor vs normal tissue immune distinctions, and associations between invasion programs and recurrence) bolster the idea that baseline tissue programs and immune composition stratify therapy outcomes and recurrence risk, and motivate measuring both immune architecture and HOXD13 status. 4) Why I chose the therapy_response endpoint: BCG efficacy is the actionable clinical outcome directly tied to the mechanistic question (how local tissue and immune state determine response to immunotherapy); therapy_response is therefore the most specific and experimentally tractable endpoint for this hypothesis. 5) What I deliberately EXCLUDED and why: I excluded cancer types/edges that are not bladder-specific (e.g., AdCC invasion and PSA/prostate-specific recurrence edges) from the core hypothesis because they provide only distal analogies; I also excluded adding additional genes as anchors because Tier 1 provided only HOXD13 as a molecular anchor and the Focusing Rule restricts the hypothesis to ≤2 anchor genes. Detailed molecular immune markers (e.g., specific myeloid/lymphoid surface genes) were not invented or added because they were not supplied by the Ontologist and would violate the constraint to avoid inventing features. Secondary modifiers capture the other relevant cell types and clinical measures for downstream analysis?

Working Hypothesis

HOXD13 upregulation in tumor-adjacent bladder tissue drives a stromal remodeling program that creates an ECM-rich, TGF-β–driven microenvironment favoring myeloid-dominant inflammation and suppressing TLS-like lymphoid organogenesis. This remodeled niche impairs local activation of adaptive immunity in response to intravesical BCG, reducing TLS maturation and BCG-induced T cell help, thereby diminishing therapeutic efficacy. Predictions: (a) HOXD13-high adjacent stroma exhibits increased fibrosis, collagen deposition, and matrix remodeling gene signatures with reduced lymphoid chemokines (e.g., CXCL13, CCL19, CCL21) and TLS-associated markers; (b) HOXD13-high, TLS-poor tumors show weaker BCG responses clinically; (c) manipulating HOXD13 levels in bladder stromal cells alters the balance of TLS versus myeloid infiltration in response to BCG in preclinical models; (d) restoring TLS-promoting signals or counteracting HOXD13-driven remodeling improves BCG efficacy. The hypothesis remains testable across three tiers: human observational data, mechanistic animal models with HOXD13 manipulation and BCG exposure, and in vitro stromal-immune interaction assays.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	HOXD13
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

HOXD13: HOXD13 upregulation in tumor-adjacent tissue) as the molecular anchor that could plausibly drive stromal/tissue remodeling and thereby modulate immune organization.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	myeloid_cells	Supporting modifier from knowledge graph
Secondary	lymphoid_cells	Supporting modifier from knowledge graph
Secondary	BCG	Supporting modifier from knowledge graph
Secondary	tumor_invasion	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- myeloid_cells: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- lymphoid_cells: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.